Technology Stack

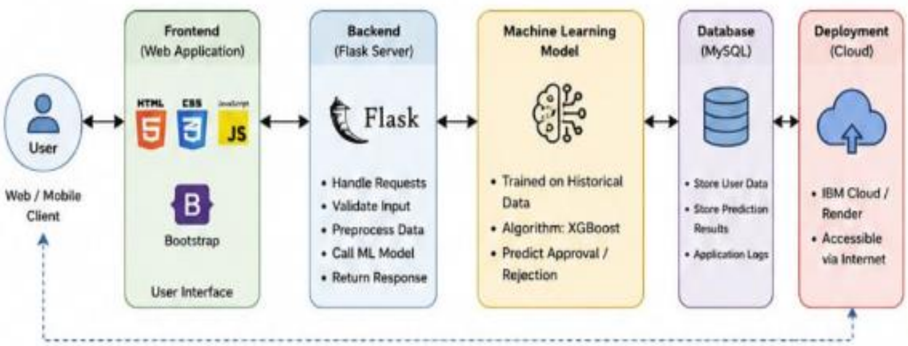
Date	26-june-2026
Team ID	SWUID2026223535
Project Name	Credit Card Approval Prediction
Maximum Marks	2 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the tablet 1 & table 2

Example: Credit Card Approval Prediction System

Reference: https://developer.ibm.com/patterns/ai-powered-credit-risk-assessment/



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

Table-1: Technology Stack (Components)

S.No	Component	Description	Technology
1.	User Interface	How user interacts with application e.g. Web UI, Mobile App, Chatbot etc.	HTML, CSS, JavaScript, Bootstrap (Responsive Web UI)
2.	Application Logic-1	Logic for a process in the application (Handle user input, routing, validation)	Python (Flask)
3.	Application Logic-2	Logic for data validation and preprocessing	Pandas, NumPy, Scikit-learn
4.	Application Logic-3	Logic for prediction using trained ML model	XGBoost Classifier
5.	Database	Data Type, Configurations etc.	MySQL
6.	Model Storage	Store trained model and scaler objects	Pickle (.pkl files)
7.	File Storage	File storage requirements (logs, reports etc.)	Local / Cloud Storage
8.	External API-1	Purpose of External API used in the application (Credit Score API)	Credit Score API (Third Party)
9.	External API-2	Purpose of External API used in the application (Email / SMS Notifications)	SendGrid API / Twilio API
10.	Machine Learning Model	Purpose of Machine Learning Model	XGBoost (Trained Model)
11.	Infrastructure (Server / Cloud)	Application Deployment on Local System / Cloud Local Server Configuration / Cloud Configuration	Render / IBM Cloud / Railway

Table-2: Application Characteristics (Part-1)

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	List the open-source frameworks used	Flask, Bootstrap, Scikit-learn, XGBoost
2.	Security Implementations	List all the security / access controls implemented	Input Validation, SQL Injection Protection, CORS, HTTPS
3.	Scalable Architecture	Justify the scalability of architecture (3 – tier, Micro-services, Modular Monolith etc.)	3-Tier Architecture (Presentation, Application, Data)
4.	Availability	Justify the availability of application (e.g. use of load balancers, distributed servers etc.)	Cloud Deployment (Render / IBM Cloud) with 99.9% uptime

Table-2: Application Characteristics (Part-2)

S.No	Characteristics	Description	Technology
5.	Performance	Design consideration for the performances of the application (number of requests per sec, use of Cache, use of CDN's) etc.	Caching (Response Cache), Optimized ML Model, Efficient Queries, CDN (If required)

References:

- https://scikit-learn.org/stable/